

Discrete Math — Monday Test

Notes + What to Practice

Quick notes for each topic, followed by exactly which questions to do (numbers match the Practice Problems / Solutions PDFs). Read the notes, then immediately do the linked questions — don't just read straight through.

Tuesday

§1.1 Propositional Logic

- A **proposition** is a statement that's definitely True or False.
- Connectives: $\neg$ (not), $\wedge$ (and), $\vee$ (or), $\rightarrow$ (implies), $\leftrightarrow$ (iff).
- $p \rightarrow q$ is only **False** when p is True and q is False. Every other combination is True (this trips people up — a false hypothesis makes the whole conditional true).
- **Converse**: $q \rightarrow p$. **Inverse**: $\neg p \rightarrow \neg q$. **Contrapositive**: $\neg q \rightarrow \neg p$ — always logically equivalent to the original.

Now do: Q1, Q2, Q3

§1.3 Propositional Equivalences

- **Tautology** = always True. **Contradiction** = always False. **Contingency** = depends.
- The single most useful identity: $p \rightarrow q \equiv \neg p \vee q$. Convert every implication to this form before applying De Morgan's.
- Named laws to recognize instantly: Identity, Domination, Idempotent, Double Negation, Commutative, Associative, Distributive, **De Morgan's** ($\neg(p \wedge q) \equiv \neg p \vee \neg q$, and the $\vee$ version), Absorption, Negation.
- **Satisfiable** = at least one truth assignment makes it True. **Unsatisfiable** = no assignment ever makes it True.

Now do: Q4, Q5, Q6, Q7, Q8

Wednesday

§1.4 Predicates & Quantifiers

- $\forall x P(x)$ — true for **every** value in the domain.
- $\exists x P(x)$ — true for **at least one** value.
- **The domain trap:** "Some students have visited X" over a broadened domain (all people) becomes $\exists x (S(x) \wedge P(x))$. "Every student has done X" becomes $\forall x (S(x) \rightarrow P(x))$ — an implication, *not* $\forall x P(x)$.
- Rule of thumb: $\exists$ pairs naturally with $\wedge$; $\forall$ pairs naturally with $\rightarrow$.
- De Morgan's for quantifiers: $\neg \forall x P(x) \equiv \exists x \neg P(x)$, and $\neg \exists x P(x) \equiv \forall x \neg P(x)$.

Now do: Q9, Q10, Q11, Q12, Q13

Thursday

§1.5 Nested Quantifiers

- **Order matters.** $\forall x \exists y (x + y = 5)$ is true (for every x you can find a matching y). $\exists y \forall x (x + y = 5)$ is false (no single y works for every x). $\forall \exists \neq \exists \forall$ in general.
- To translate, work outside-in: read the outermost quantifier first, then the next, then the predicate.
- To negate a nested-quantifier statement, push the negation through one quantifier at a time using De Morgan's, flipping $\forall \leftrightarrow \exists$ each time, until it sits directly on the predicate.

Now do: Q14, Q15, Q16, Q17

§1.6 Rules of Inference — start

- An argument is **valid** if the conclusion must be true whenever all premises are true — check this by seeing if the argument form is a tautology.
- Core rules: **Modus Ponens** ($p \rightarrow q, p \therefore q$), **Modus Tollens** ($p \rightarrow q, \neg q \therefore \neg p$), **Hypothetical Syllogism** ($p \rightarrow q, q \rightarrow r \therefore p \rightarrow r$), **Disjunctive Syllogism** ($p \vee q, \neg p \therefore q$).

Now do: Q18, Q19

Friday

§1.6 Rules of Inference — finish, & Fallacies

- More rules: **Addition** ($p \therefore p \vee q$), **Simplification** ($p \wedge q \therefore p$), **Conjunction** ($p, q \therefore p \wedge q$), **Resolution** ($p \vee q, \neg p \vee r \therefore q \vee r$).
- Quantified versions: **Universal Instantiation** ($\forall x P(x) \therefore P(c)$), **Existential Instantiation** ($\exists x P(x) \therefore P(c)$ for *some* specific c), and their **Generalization** counterparts running the other way.
- Multi-step proofs = chain these rules one line at a time, naming the rule used on each line.
- **Classic fallacies:** affirming the consequent ($p \rightarrow q, q \therefore p$ — INVALID) and denying the antecedent ($p \rightarrow q, \neg p \therefore \neg q$ — INVALID). Test with a truth table: if you can make premises True and conclusion False, it's a fallacy.

Now do: Q20, Q21, Q22

Saturday

§1.7 Introduction to Proofs

- **Direct proof:** assume p , derive q step by step through algebra/logic.
- **Proof by contraposition:** assume $\neg q$, derive $\neg p$ instead — use when the direct route gets messy.
- **Proof by contradiction:** assume the hypothesis *and* the negation of the conclusion, derive something impossible.
- **Counterexample:** to disprove a $\forall$ statement, find just one case where it fails.

Now do: Q23, Q24, Q25, Q26, Q27

§1.8 Proof Methods and Strategy

- **Proof by cases:** split into an exhaustive set of scenarios, prove each one separately.
- **Proof by exhaustion:** when the domain is small and finite, just check every value.
- **Existence proofs:** *constructive* = you exhibit the object explicitly; *nonconstructive* = you prove something must exist without identifying it.
- Watch for the classic algebra trap: never divide both sides by an expression that could be zero (that's how "1=2" fake proofs sneak in).

Now do: Q28, Q29, Q30, Q31, Q32

Sunday

§5.1 Mathematical Induction

- Three-part skeleton, every time:
 1. **Base case:** verify $P(n_0)$ (usually $n = 1$ or $n = 0$).
 2. **Inductive hypothesis:** assume $P(k)$ is true for some arbitrary k .
 3. **Inductive step:** using that assumption, prove $P(k + 1)$.
- The algebra move: add the $(k + 1)$ -th term to both sides of the $P(k)$ formula, then factor/simplify the right side until it matches the target shape for $n = k + 1$. That algebra *is* the proof.
- **Inequality induction** (e.g. $2^n > n^2$) uses the same skeleton but you often need an extra inequality step (like $2k^2 > (k + 1)^2$) inside the inductive step.
- **Spot-the-error questions** (like the all-horses-one-color fallacy) test whether the inductive step *actually* holds for every k , especially the very first jump from the base case.

Now do: Q33, Q34, Q35, Q36, Q37, Q38

Sunday, last 40 min: pick 2–3 questions at random from earlier in the week (not ones you just redid) and solve them cold, timed, no notes. That’s your best readiness check before Monday.